# Skill, risk and luck in feedback-guessing games

The Lexodle project

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**Abstract**

We define skill, risk and luck for a guess in a word game with colour feedback (Wordle and variants). A guess partitions the words still possible by the feedback they would give. The answer lies in one class; its size is the *pool*. *Skill* is the log-reduction of the expected pool, as a fraction of the same quantity for the best real guess. *Luck* is the probability of a larger pool minus the probability of a smaller one. It lies strictly in $(-100, 100)$ percent, has mean zero for every guess, and its increments over a game are martingale differences, so luck is uncorrelated with skill. The two simpler luck definitions are unbounded or biased. The log-reduction of the candidate set equals skill plus risk plus luck exactly, where risk is the gap between Shannon and collision entropy. *Daring* measures how far the player went for the whole word instead of the clues: it is zero for a guess that cannot be the answer, and otherwise grows with what the attempt cost in skill, how unlikely the word was, and how many tries were left. We extend the definitions to games with several boards, give an exact algorithm, and report experiments on an English five-letter list ($N = 4{,}994$) and simulated players of several habits. The best first guess leaves 114.4 words on average. CRANE scores 81.6 (97.6 on a linear scale, which puts most openings near 100). A bound on the best conceivable split from the number of feedback patterns is too loose to use. We list open problems.

## 1 Introduction

In Wordle-like games the player guesses a word. Each letter is marked as in the right place, elsewhere in the word, or absent. After the game two questions arise. Skill: given what was known, how good was the guess? Luck: given the guess, how favourable was the feedback? A score that mixes the two is uninformative.

Entropy-based solvers choose the guess that maximises the expected information in the feedback [1, 6], as for the code game Mastermind [2, 3]. Exact methods give optimal strategies for the original word list [4]. For generalised word lists, deciding whether a win within a given number of guesses can be guaranteed is NP-hard [5]. A newspaper review tool reports skill and luck per guess [7]; we know of no published formula. This paper does not address how to play. It defines scores to show a player after the game. The requirements are:

(R1) **Knowledge only.** Neither score depends on the answer except through what the player saw.

(R2) **Bounded.** Skill runs from 0 (the guess tells nothing) to 100 (as good as any real guess). Luck is symmetric about 0 and bounded.

(R3) **Fair luck.** Luck has mean zero and is uncorrelated with skill.

(R4) **One plot.** Both scores can be read from a single plot of the outcomes the guess could have had.

(R5) **Accounting.** The reduction of the candidate set equals a skill term plus a risk term plus a luck term, exactly (Section 5).

Section 2 fixes the setting, Section 3 defines skill, Section 4 luck, Section 5 the accounting, Section 6 daring, Section 7 several boards, Section 8 the algorithm, Section 9 the experiments, Section 10 the open problems. Code and scripts are listed in Section 11.

## 2 Setting

Let $\Omega$ be the finite set of words still possible given everything the player has seen, $N = |\Omega| \geq 1$, and $\mathcal{G}$ the finite set of admissible guesses. Assume $\Omega \subseteq \mathcal{G}$. Let $\Phi$ be the set of feedback patterns ($\{\text{grey}, \text{yellow}, \text{green}\}^n$ for $n$-letter words) and $f\colon \mathcal{G} \times \Omega \to \Phi$ the feedback function. Only two properties of $f$ are used: it is computable, and $f(g, w)$ is all green only if $g = w$.

**Definition 1** (Prior (U))**.** The answer $A$ is uniform on $\Omega$. Given the history of guesses and feedback, $A$ is uniform on the words consistent with it.

(U) is the player's view: without knowing how the game draws answers, all possible words count equally. Section 9 measures the effect of the real draw.

A guess $g \in \mathcal{G}$ partitions $\Omega$ into nonempty classes $C_\varphi = \{w \in \Omega : f(g, w) = \varphi\}$ of sizes $s_1, \ldots, s_k$, $\sum_i s_i = N$. The answer lands in one class. Its size $X_g = |C_{f(g,A)}|$ is the *pool*, the number of words still possible after the guess. If $m_x$ classes have size $x$,

$$\mathbb{P}(X_g = x) = \frac{x\, m_x}{N}. \tag{1}$$

Pools are size-biased: a large class is more likely to contain the answer.

## 3 Skill

**Definition 2** (Expected pool)**.** $E(g) = \mathbb{E}[X_g] = \dfrac{1}{N} \sum_{i=1}^{k} s_i^2$.

**Definition 3** (Best real guess, skill)**.** $E^* = \min_{g \in \mathcal{G}} E(g)$. For $N \geq 2$ the *skill* of $g \in \mathcal{G}$ is

$$S(g) = \frac{\ln\bigl(N/E(g)\bigr)}{\ln\bigl(N/E^*\bigr)} \in [0, 1],$$

reported as a percentage. For $N = 1$, $S(g) = 1$ if $g$ is the remaining word and 0 otherwise.

The logarithm is used because a guess cuts the candidate set by a factor. On the linear scale $(N - E)/(N - E^*)$ most sensible guesses score near 100 (Section 9). The linear score has the same endpoints and the same ordering of guesses.

**Lemma 1.** *Let the partition of $g$ have $k$ classes.*

1. $\max\{1, N/k\} \le E(g) \le N$; $E(g) = N$ *iff* $k = 1$; $E(g) = 1$ *iff* $k = N$; $E(g) = N/k$ *iff all classes have equal size.*

2. *If the partition of* $g'$ *properly refines that of* $g$*, then* $E(g') < E(g)$.

*Proof.* (1) $1 \le s_i \le N$ gives $E \ge 1$, with equality iff every $s_i = 1$, and $s_i^2 \le N s_i$ gives $E \le N$, with equality iff $k = 1$. By Cauchy–Schwarz $\sum s_i^2 \ge (\sum s_i)^2/k = N^2/k$, with equality iff the $s_i$ are equal. (2) Splitting a class of size $a_1 + \cdots + a_r$, $r \ge 2$, replaces $(\sum a_j)^2$ by $\sum a_j^2$, which is smaller because the cross terms are positive. □

**Proposition 2.** *Let* $N \ge 2$ *and* $\Omega \subseteq \mathcal{G}$*. Then* $E^* < N$*, so* $S$ *is defined, and*

1. $0 \le S(g) \le 1$ *for every* $g \in \mathcal{G}$; $S(g) = 1$ *iff* $E(g) = E^*$; $S(g) = 0$ *iff* $g$ *does not divide* $\Omega$;

2. $S$ *is strictly decreasing in* $E(g)$ *and affine in* $\ln E(g)$*: halving the expected pool adds* $\ln 2/\ln(N/E^*)$*. If the partition of* $g'$ *properly refines that of* $g$*, then* $S(g') > S(g)$;

3. $S(g)$ *is a function of* $(\Omega, g, \mathcal{G})$ *alone. It does not depend on the answer.*

*Proof.* Take $w \in \Omega \subseteq \mathcal{G}$. Only $w$ gives all green against $w$, so $\{w\}$ is a class of $w$'s partition. For $N \ge 2$ the partition is nontrivial, so $E(w) < N$ by Lemma 1 and $E^* < N$. The denominator is positive. (1) follows from $E^* \le E(g) \le N$ and the lemma, (2) from the formula and Lemma 1(2), (3) because $E(g)$ and $E^*$ are computed from the classes of $\Omega$. □

*Remark* 1 (Entropy). Let $p_i = s_i/N$. Then $\ln(N/E) = -\ln \sum p_i^2 = H_2$, the Rényi entropy of order 2 [9]; $e^{H_2}$ is the inverse Simpson index, the Hill number of order 2 [8, 10]. Hence $S(g) = H_2(g)/\max_{g'} H_2(g')$. The Shannon entropy $H = -\sum p_i \ln p_i = \ln N - \mathbb{E}[\ln X_g]$ [1] is the expected log-reduction of the candidate set, and $H \ge H_2$ by Jensen. $H_2$ is used because it is a function of $E$, the expected number of words left. Other orders of Rényi entropy give other scales (Section 9.4).

*Remark* 2 (Why the best real guess). A bound that needs no search is $e = N/\min(N, 3^n)$, the most even split into $\min(N, 3^n)$ groups. Section 9 tests it and a sharper bound from the clues. Both are too loose. The best real word makes about two thirds of the groups the bound allows when many words remain, and rarely separates every word when few remain. $E^*$ is therefore the actual minimum over real guesses.

## 4 Luck

Let $q_x = \mathbb{P}(X_g = x) = x\, m_x/N$.

**Definition 4** (Luck)**.** For $x$ in the support of $X = X_g$,

$$\Lambda_g(x) = \mathbb{P}(X > x) - \mathbb{P}(X < x).$$

The luck of an outcome is $\Lambda_g(X_g)$, reported as a percentage.

It is the probability of a larger pool minus the probability of a smaller one: negative if the outcome is worse than most, positive if better, zero at the median.

**Theorem 3.** *Let* $X$ *have the law* (1) *and* $\Lambda = \Lambda_g$.

1. *(Mid-distribution form.)* $\Lambda(x) = 2\tilde{F}(x) - 1$ *with* $\tilde{F}(x) = \mathbb{P}(X > x) + \frac{1}{2}\,\mathbb{P}(X = x)$. $\Lambda(X)$ *is Cliff's delta between the realised pool and an independent copy [14].*

2. *(Bounded.)* $-1 < \Lambda(x) < 1$ *for every* $x$ *in the support, and* $\Lambda$ *is strictly decreasing in* $x$.

3. *(Fair.)* $\mathbb{E}[\Lambda(X)] = 0$ *for every guess.*

4. *(Spread.)* $\operatorname{Var}\Lambda(X) = \dfrac{1 - \sum_x q_x^3}{3} < \dfrac{1}{3}$.

5. *(Invariance.)* $\Lambda(x)$ *is unchanged if all class sizes are multiplied by the same positive factor. Equal pools have equal luck.*

*Proof.* (1) $\mathbb{P}(X < x) = 1 - \mathbb{P}(X > x) - \mathbb{P}(X = x)$, so $\Lambda(x) = 2\,\mathbb{P}(X > x) + \mathbb{P}(X = x) - 1 = 2\tilde{F}(x) - 1$. For an independent copy $X'$, $\Lambda(x) = \mathbb{E}[\mathbf{1}\{X' > x\} - \mathbf{1}\{X' < x\}]$, which is Cliff's delta.

(2) $\mathbb{P}(X > x) + \mathbb{P}(X < x) = 1 - q_x < 1$ and both terms are non-negative, so $|\Lambda(x)| < 1$. If $x < y$ are in the support, $\mathbb{P}(X > x) \geq \mathbb{P}(X > y) + q_y$ and $\mathbb{P}(X < x) \leq \mathbb{P}(X < y) - q_x$, so $\Lambda(x) > \Lambda(y)$.

(3) For independent identically distributed $X, X'$, $\mathbb{E}[\Lambda(X)] = \mathbb{P}(X' > X) - \mathbb{P}(X' < X) = 0$ by exchangeability.

(4) Order the distinct pool sizes $x_1 > \cdots > x_r$ with masses $q_i$ and $c_i = \sum_{j<i} q_j = \mathbb{P}(X > x_i)$, so $\tilde{F}(x_i) = c_i + q_i/2$. The sum $\sum_i [(c_i + q_i)^3 - c_i^3] = 1$ telescopes, so

$$\sum_i \bigl(3c_i^2 q_i + 3c_i q_i^2 + q_i^3\bigr) = 1, \quad \text{hence} \quad \sum_i q_i c_i^2 + \sum_i c_i q_i^2 = \frac{1 - \sum_i q_i^3}{3}.$$

Then $\mathbb{E}[\tilde{F}^2] = \sum_i q_i (c_i + q_i/2)^2 = \sum_i q_i c_i^2 + \sum_i c_i q_i^2 + \frac{1}{4}\sum_i q_i^3 = \frac{1}{3} - \frac{1}{12}\sum_i q_i^3$. By (3), $\mathbb{E}[\tilde{F}] = \frac{1}{2}$, so $\operatorname{Var}\tilde{F} = \frac{1}{12}(1 - \sum_i q_i^3)$ and $\operatorname{Var}\Lambda = 4\operatorname{Var}\tilde{F}$.

(5) Multiplying every $s_i$ by $c > 0$ multiplies $N$ and each numerator $x\,m_x$ by $c$, so each $q_x$ is unchanged. □

*Remark* 3. $\Lambda$ is the mid-distribution transform of the pool [11, 12], a discrete analogue of the probability integral transform [13]. Giving tied pools their mid-rank makes the mean exactly zero. If columns of a plot are as wide as their chance, the horizontal middle is the median outcome and $\Lambda$ is the position of the landed column relative to that middle (Figure 1).

### 4.1 Luck is uncorrelated with skill

Let $\mathcal{F}_{t-1}$ be the history before guess $t$. The guess $g_t$ is a function of $\mathcal{F}_{t-1}$ and possibly independent randomness. Let $S_t$ and $\Lambda_t = \Lambda_{g_t}(X_{g_t})$ be its skill and luck.

**Theorem 4.** *Under (U),* $\mathbb{E}[\Lambda_t \mid \mathcal{F}_{t-1}] = 0$. *Hence* $\mathbb{E}\,\Lambda_t = 0$, $\operatorname{Cov}(\Lambda_t, S_s) = 0$ *for* $s \leq t$, $\operatorname{Cov}(\Lambda_s, \Lambda_t) = 0$ *for* $s \neq t$, *and* $\sum_{s \leq t} \Lambda_s$ *is a martingale.*

*Proof.* Given $\mathcal{F}_{t-1}$ the answer is uniform on the consistent words $\Omega_t$ and $g_t$ is fixed, so Theorem 3(3) applies conditionally. $S_s$ ($s \leq t$) and $\Lambda_s$ ($s < t$) are $\mathcal{F}_{t-1}$-measurable, so $\operatorname{Cov}(\Lambda_t, S_s) = \mathbb{E}[S_s\,\mathbb{E}[\Lambda_t \mid \mathcal{F}_{t-1}]] - \mathbb{E}\,S_s\;\mathbb{E}\,\Lambda_t = 0$, and likewise for $\Lambda_s\Lambda_t$. □

This is (R3).

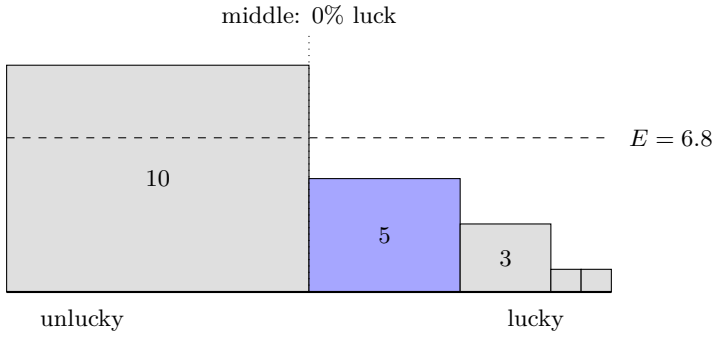



Figure 1: Example 1: twenty words in groups of $10, 5, 3, 1, 1$. Column width is chance, height is pool size. The dashed line is $E = 6.8$. The shaded group of 5 has luck +25%: chance of a larger pool (10/20) minus chance of a smaller one (5/20).

### 4.2 Alternative luck definitions

Two definitions scale the difference between the landed pool $a$ and the expected pool $E$.

**Proposition 5.** *Let $E > 1$.*

1. *The expectation-scaled luck $\Lambda^{\exp}(a) = (E-a)/(E-1)$ has mean zero and no lower bound: for one pair and $N-2$ singletons, landing in the pair gives $-(N-2)/2$.*
2. *The rescaled luck $\Lambda^{\mathrm{res}}(a) = (E-a)/(E-1)$ for $a \le E$ and $(E-a)/(a_{\max}-E)$ for $a > E$ lies in $[-1, 1]$. Its mean is $\kappa\big(\frac{1}{E-1} - \frac{1}{a_{\max}-E}\big)$, $\kappa = \mathbb{E}[(E-X)^+]$, which is nonzero unless $a_{\max} - E = E - 1$. For the partition $\{3, 1\}$ it is $-\frac{1}{2}$.*

*Proof.* (1) $\mathbb{E}\,\Lambda^{\exp} = (E - \mathbb{E}\,X)/(E-1) = 0$. For a pair and $N-2$ singletons $E = (N+2)/N$, so $E - 1 = 2/N$, $E - 2 = -(N-2)/N$, and the ratio is $-(N-2)/2$. (2) $\mathbb{E}[X - E] = 0$ gives $\mathbb{E}[(E-X)^+] = \mathbb{E}[(X-E)^+] = \kappa$. The positive part of $\Lambda^{\mathrm{res}}$ is scaled by $1/(E-1)$ and the negative part by $1/(a_{\max} - E)$. For $\{3, 1\}$: $N = 4$, $E = \frac{5}{2}$, $a_{\max} = 3$. Landing in the 3 (chance $\frac{3}{4}$) gives $-1$, landing in the 1 gives $+1$, mean $-\frac{1}{2}$. □

The first violates (R2), the second (R3): its bias mixes guess quality into luck. Section 9 gives means from +19% to −60% on real positions.

### 4.3 The last try

On the last guess only a word that fits the clues can win. Let $n$ words fit. A guess that is one of them wins with chance $1/n$. Any other guess cannot win.

**Proposition 6.** *Let $Y = \mathbf{1}\{\mathit{win}\}$ and $\pi = \mathbb{P}(\mathit{win}) \in \{1/n, 0\}$. The last-try luck $\Lambda = Y - \pi$ has mean zero for every guess, takes the values $1-\pi$ and $-\pi$, and lies in $(-1, 1)$ for $n \ge 2$. It is 0 for a guess that cannot win; that loss belongs to skill. With three words fitting, a hit is +67% and a miss −33%.*

*Proof.* $\mathbb{E}\,Y = \pi$, so $\mathbb{E}[Y - \pi] = 0$. The range follows from $0 \le \pi \le \frac{1}{2}$ for $n \ge 2$. □

**Example 1.** Twenty words are split into groups of $10, 5, 3, 1, 1$. Then $E = (100 + 25 + 9 + 1 + 1)/20 = 6.8$. If no real guess beats singletons ($E^* = 1$), the skill is $\ln(20/6.8)/\ln 20 = 36.0\%$ ($(20 - 6.8)/19 = 69.5\%$ on the linear scale). Luck by landing: in the 10, $0 - \frac{10}{20} = -50\%$; in the 5, $\frac{10}{20} - \frac{5}{20} = +25\%$; in the 3, $\frac{15}{20} - \frac{2}{20} = +65\%$; in a 1, $\frac{18}{20} - 0 = +90\%$ (the two singletons tie and neither counts against the other). The chance-weighted mean is $\frac{10}{20}(-.5) + \frac{5}{20}(.25) + \frac{3}{20}(.65) + \frac{2}{20}(.9) = 0$.

# 5 Accounting: skill, risk and luck

Skill and luck have different scales, so they do not add. Their underlying quantities do. Let $\rho = \ln(N/A)$ be the *total reduction* of the candidate set (nats; divide by $\ln 2$ for bits): $A$ words remain of $N$. With $X = X_g$ and $H = -\sum_i p_i \ln p_i$,

$$\underbrace{H_2}_{\text{skill}} + \underbrace{H - H_2}_{\text{risk}} + \underbrace{\rho - H}_{\text{luck}} = \rho, \qquad H_2 = \ln \frac{N}{E}.$$

**Proposition 7** (Accounting)**.** *For every guess and outcome the three terms sum to $\rho$. Moreover*

1. *the skill term $H_2 = \ln(N/E) \geq 0$ is the numerator of $S$;*
2. *the risk term $H - H_2 = \ln \mathbb{E}[X] - \mathbb{E}[\ln X] \geq 0$, with equality iff all classes have equal size;*
3. *the luck term $Z = \rho - H = \mathbb{E}[\ln X] - \ln A$ has $\mathbb{E}[Z] = 0$ and $\operatorname{Var} Z = \operatorname{Var}(\ln X)$, and under the hypotheses of Theorem 4 it is a martingale difference.*

*Proof.* $\mathbb{E}[\rho] = \mathbb{E}[\ln(N/X)] = \ln N - \mathbb{E} \ln X = H$ and $H_2 = \ln N - \ln E$. Hence $H - H_2 = \ln \mathbb{E}[X] - \mathbb{E}[\ln X] \geq 0$ by Jensen's inequality, with equality iff $X$ is constant, which by (1) holds iff all class sizes are equal. The luck term is $\rho - \mathbb{E}\,\rho$: mean zero, variance $\operatorname{Var} \rho = \operatorname{Var}(\ln X)$. The conditional statement follows as in Theorem 4, applied to $\ln X$. □

*Remark* 4 (Risk). The risk term is the convexity gap between the average of the log and the log of the average. An even split has $H = H_2$ and zero risk. One large class and several small ones has a large gap: low expected reduction, high payoff if the answer is in a small class. Take 100 words in ten classes of ten, against one class of 91 and nine singletons. The first has $H_2 = H = 3.32$ bits and risk 0. The second has skill 0.27, risk 0.45 and expected reduction 0.72 bits. Landing in a singleton (chance 9%) reduces by 6.64 bits $= 0.27 + 0.45 + 5.92$. Landing in the 91 (chance 91%) reduces by $0.14 = 0.27 + 0.45 - 0.59$.

*Remark* 5 (Two luck scores). $Z = \mathbb{E}[\ln X] - \ln A$ and the percentile luck $\Lambda$ are both strictly decreasing in the landed pool $A$ and both have mean zero, so they rank outcomes identically. $Z$ has the units of skill and risk, which makes the accounting exact, and is bounded by $\pm \ln N$. $\Lambda$ is the mid-rank of the pool and is bounded by $\pm 1$.

Table 1 gives the terms for some first guesses.

# 6 Daring

Risk is a property of a guess. It does not say what the player was trying to do. A guess that could be the answer is an attempt at the word; one that cannot is a probe, chosen for what

| guess | skill $H_2$ | risk $H - H_2$ | expected $H$ | sd of $\rho$ |
|---|---|---|---|---|
| tares | 5.45 | 0.77 | 6.22 | 1.71 |
| slate | 5.18 | 0.78 | 5.96 | 1.78 |
| crane | 4.44 | 0.97 | 5.42 | 2.02 |
| abbey | 3.18 | 0.70 | 3.88 | 1.91 |
| queue | 2.04 | 0.77 | 2.81 | 1.99 |
| mamma | 1.57 | 0.84 | 2.41 | 2.18 |

Table 1: The terms for first guesses (bits), $N = 4{,}994$. For CRANE, landing in a 641-word class reduces by 2.96 bits ($= 4.44 + 0.97 - 2.46$), in a 23-word class by 7.76 ($= 4.44 + 0.97 + 2.34$), in a single word by 12.29 ($= 4.44 + 0.97 + 6.87$).

it tells. *Daring* measures how much a guess was an attempt at the whole word rather than a way of narrowing the pool. Fix a position with $\tau \geq 1$ tries left, this one included, a set $\Omega$ of words still possible, and a weight $w(\omega)$ for each (its frequency, which is how the game draws the answer). Let $S(g) \in [0, 100]$ be the skill of $g$, and

$$p(g) = \frac{w(g)}{\sum_{\omega \in \Omega} w(\omega)} \quad \text{if } g \in \Omega, \qquad p(g) = 0 \quad \text{otherwise,}$$

the chance, given the clues, that $g$ is the answer.

**Definition 5** (Daring)**.** The daring of a guess $g$ is

$$D(g) = \mathbf{1}[g \in \Omega] \left(1 - \frac{S(g)}{100}\right) \left(1 - p(g)\right) \left(1 - \frac{1}{\tau}\right).$$

It is quoted in percent, and is undefined when only one word is possible.

The four factors say, in turn: the guess is a try at the word; it gave up some of the splitting the best guess would have done; the word was unlikely; and there were tries to spare. With one try left going for the word is the only play, so $D = 0$. A guess that is both the best split and a possible answer costs nothing, so $D = 0$ too.

**Proposition 8.** *1. $D(g) \in [0, 1]$, and it depends on the position, the choice of $g$ and $\tau$ only, not on the answer, so (R1) holds.*

*2. $D(g) = 0$ if $g \notin \Omega$, if $\tau = 1$, if $S(g) = 100$, or if $p(g) = 1$.*

*3. $D$ is non-increasing in $S$ and in $p$ and non-decreasing in $\tau$, the others fixed. It is largest, $1 - 1/\tau$, for a possible answer of skill 0 and chance 0.*

*4. $D$ does not involve risk, so the accounting of Proposition 7 is unaffected.*

*Proof.* All four are read off the product: each factor lies in $[0, 1]$, is monotone in the stated variable, and the first, third and fourth do not mention the answer. □

*Remark* 6. Daring is reported beside the accounting and is not a term of it. An earlier definition counted, among the guesses of higher skill, the share that were less risky; it mostly measured how far a guess was from the best one, so any player who was not optimal looked daring, and it

ignored the tries left. It is replaced because the question a player means by "bold" is whether they went for the word. Skill is relative to the language and the word length, so $D$ is too, through $S$. The weights are those of the app's word list; in a quartet or relay race the weights are not used and every possible word counts as equally likely.

# 7 Several boards

Two variants hide several words. In a *quartet* one guess is played on every unsolved board at once. In a *relay race* boards are played in turn, and every earlier guess is applied to a board when it opens.

A guess is made for live boards $b = 1, \dots, B$ with possible sets $\Omega_b$ of sizes $N_b$. Skill $S_b$ and luck $\Lambda_b$ are computed per board as above, with $E_b^*$ the best real guess for that board. Let $w_b = \log_2 \max\{N_b, 2\}$, which depends only on the history, and

$$S = \frac{\sum_b w_b S_b}{\sum_b w_b}, \qquad \Lambda = \frac{\sum_b w_b \Lambda_b}{\sum_b w_b}.$$

**Proposition 9.** *$S \in [0,1]$ and $-1 < \Lambda < 1$. If, given the history, each answer is uniform on its board's candidates, then $\mathbb{E}[\Lambda \mid \mathcal{F}_{t-1}] = 0$ and Theorem 4 holds for the combined luck. Independence between boards is not required.*

*Proof.* Convex combinations stay in $[0,1]$ and $(-1,1)$. The weights are $\mathcal{F}_{t-1}$-measurable, so $\mathbb{E}[\Lambda \mid \mathcal{F}_{t-1}] = \sum_b w_b \, \mathbb{E}[\Lambda_b \mid \mathcal{F}_{t-1}] / \sum_b w_b = 0$ by Theorem 3(3) on each board. The remainder of Theorem 4's proof uses only this. □

*Remark* 7 (Weights). $\log_2 N_b$ is the number of bits still to find on board $b$. A board with one word left has weight one: playing that word is rewarded and missing it is penalised; its skill is 1 if the guess is that word and 0 otherwise. Uniform weights would let a nearly solved board dominate. The weights are a design choice.

*Remark* 8 (Head start). In a relay race a guess on the active board narrows boards not yet open. This credit is not scored. It is reported: the *head start* of a board is the number of words possible on it when it opens, with all earlier guesses applied, against the size of the list. An earlier guess can solve a board as it opens. Assigning this credit to the earlier guesses is open.

# 8 Computation

Feedback for an $n$-letter guess is packed into an integer in $\{0, \dots, 3^n - 1\}$. For a guess $g$, class sizes are counted in an array indexed by pattern, reset by revisiting only the patterns touched. $E(g)$ costs one pass, $O(nN)$, with no allocation. Skill needs $E^* = \min_g E(g)$, a scan of all $|\mathcal{G}|$ guesses: $O(nN|\mathcal{G}|)$, about $1.2 \times 10^8$ operations for a first guess on $N = |\mathcal{G}| = 4{,}994$ five-letter words, and fewer as $N$ falls. The scan also gives the skill of every guess, so the rank of the player's guess is free. For several boards, boards sharing the same candidate list (all of them at the start) are scored once. A truncated scan gives $E^*$ too large and skill too high; the app scans every guess.

Luck needs the class sizes of the player's guess only: sort the distinct sizes and accumulate the masses above and below, $O(k \log k)$ for $k$ classes.

| guess | rank | $E$ | groups | largest | skill | linear |
|---|---|---|---|---|---|---|
| tares | 1 | 114.4 | 184 | 350 | 100.0 | 100.0 |
| slate | 24 | 138.0 | 174 | 355 | 95.0 | 99.5 |
| trace | 534 | 219.8 | 155 | 625 | 82.7 | 97.8 |
| crane | 628 | 229.3 | 155 | 641 | 81.6 | 97.6 |
| abbey | 3786 | 550.9 | 78 | 1069 | 58.4 | 91.1 |
| queue | 4907 | 1218.5 | 40 | 2063 | 37.4 | 77.4 |
| mamma | 4973 | 1681.5 | 49 | 2693 | 28.8 | 67.9 |
| xxxix | 4994 | 2828.0 | 21 | 3598 | 15.1 | 44.4 |

Table 2: First guesses ranked by $E$ among all 4,994 words.

| | | | largest pool | | | mean over answers | |
|---|---|---|---|---|---|---|---|
| guess | $E$ | largest | exp. | resc. | chance | resc. | chance |
| tares | 114.4 | 350 | −208% | −100% | −93% | +18.8% | 0.0% |
| crane | 229.3 | 641 | −180% | −100% | −87% | +18.1% | 0.0% |
| abbey | 550.9 | 1069 | −94% | −100% | −79% | −1.8% | 0.0% |
| queue | 1218.5 | 2063 | −69% | −100% | −59% | −12.7% | 0.0% |
| mamma | 1681.5 | 2693 | −60% | −100% | −46% | −21.5% | 0.0% |

Table 3: Luck of landing in the largest pool (expectation-scaled, rescaled, chance-weighted) and mean luck over all answers. The expectation-scaled mean is zero by construction.

# 9 Experiments

All figures use the app's English five-letter list ($N = 4{,}994$; every word is a legal guess and a possible answer). Scripts are in `docs/paper/experiments`; outputs in its `results` directory.

## 9.1 First guesses

Table 2 gives $E$, the number of groups, the largest group and the skill of some openings, with the linear skill $(N - E)/(N - E^*)$ for comparison. The best first guess, TARES, leaves 114.4 words on average.

Over all 4,994 words as first guesses, the median linear skill is 94.3 and 81.1% score above 90 (42.7% above 95). On the logarithmic scale the median is 67.4 and 2.7% score above 90 (0.5% above 95). The linear scale discriminates little among sensible openings.

## 9.2 Luck definitions

Table 3 gives the luck of landing in the largest pool and the mean luck over all answers, for each definition of Section 4.2. The expectation-scaled luck reaches −208% for the largest pool of TARES. The rescaled luck lies in $[-100, 100]$ but its mean over answers ranges from +18.8% to −21.5% on first guesses, and is −60% in a late position (five words left). The chance-weighted luck has mean 0 to the precision shown.

For CRANE, landing in the 641-word pool (chance 12.8%) gives −87%, in a 328-word pool

| valuation | opener | mean guesses | $\leq 3$ | $\leq 4$ | $\leq 5$ | $\leq 6$ |
|---|---|---|---|---|---|---|
| worst case ($\alpha = \infty$) | aloes | $4.01 \pm 0.03$ | 26.9 | 75.3 | 95.9 | 99.5 |
| collision ($\alpha = 2$) | tares | $3.85 \pm 0.03$ | 34.3 | 82.1 | 97.0 | 99.4 |
| Shannon ($\alpha = 1$) | tares | $3.83 \pm 0.03$ | 36.2 | 82.5 | 96.9 | 99.1 |
| groups ($\alpha = 0$) | tares | $3.81 \pm 0.03$ | 37.8 | 82.5 | 97.0 | 99.2 |

Table 4: Greedy player under four valuations, 1,000 games each. Shares in percent, standard error 1.2 to 1.5 points.

$-47\%$, in a 23-word pool $+71\%$, in a single word $+99\%$. Landing in a pair among $N - 2$ singletons gives $-(N-2)/N$ chance-weighted and $-(N-2)/2$ expectation-scaled: $-98\%$ and $-4{,}900\%$ at $N = 100$.

### 9.3 Simulated players

`play_skill.py` plays 60 games each for a strong player (a good opener, then the consistent word leaving the fewest words) and a casual one (a random common opener, then a random consistent word). Answers are uniform on the list. Guesses are scored on the logarithmic scale. Per guess the strong player's median is 100 (mean 90) and the casual player's 74 (mean 74). In won games the strong player has every guess above 80 in 64% and above 90 in 27%; the casual player in 0% for both. On the linear scale nearly every guess of either player would score above 90.

### 9.4 Is risk worth rewarding?

$H_2$ rewards even splits. $H$ (the risk term above) credits a split into one large and several small classes more, and counting groups more still. These are the Rényi entropies of order $\alpha = \infty$ (the largest class), 2 (the app), 1 (Shannon) and 0 (number of classes). `risk_attitude.py` plays a greedy player ranking guesses by each. The first guess is chosen over every word; later guesses among the words still possible and a sample of common words. The same 1,000 uniformly drawn answers are used for each (Table 4).

The worst-case valuation is slower (4.01 against about 3.8, five standard errors). The other three have indistinguishable mean time. The share won within three guesses rises with the weight on small classes (34.3%, 36.2%, 37.8%) with no loss within six (99.4, 99.1, 99.2%). This is consistent with a gamble paying off when few guesses are needed, but the evidence is weak: one greedy policy, a restricted candidate set, one language and length, one run. The three non-worst-case players share the opener TARES, so their differences come from later guesses.

### 9.5 Daring and playstyle: simulated players

The app reads a playstyle from a player's average skill and average daring. To set its thresholds, `app/tools/calibrate_styles.dart` plays seven simulated players through the app's own code (not the Python of the other experiments), with answers drawn by frequency from the 3,000 most common words. The players are: *best* (the guess with the smallest $E$ every turn); *near-best* (one of the three best at random); *best-fit* (the best split among the words that still fit, so always an attempt at the word); *frequent* (the most common word that still fits); *random-fit*

| player | won (%) | guesses | skill | daring |
|---|---|---|---|---|
| best | 100 | 3.6 | 100.0 | 0.0 |
| near-best | 100 | 3.7 | 99.0 | 0.3 |
| best-fit | 100 | 3.4 | 91.4 | 4.0 |
| frequent | 100 | 3.1 | 66.3 | 19.7 |
| random-fit | 97 | 4.3 | 70.7 | 21.4 |
| wild | 1 | 6.0 | 48.2 | 6.0 |
| wild-bold | 0 | 6.0 | 46.3 | 6.5 |

Table 5: Simulated players, English, five letters, 200 games each. Won within six guesses, guesses per game, and the means of skill and daring.

| skill of the guess | guesses | daring p10 | p50 | p90 |
|---|---|---|---|---|
| 0–40 | 830 | 0 | 0 | 19 |
| 40–55 | 620 | 0 | 0 | 42 |
| 55–70 | 901 | 0 | 28 | 33 |
| 70–80 | 468 | 0 | 18 | 23 |
| 80–90 | 361 | 0 | 10 | 14 |
| 90–100 | 377 | 0 | 1 | 6 |
| 100 | 1470 | 0 | 0 | 0 |

Table 6: Daring by the skill of the guess, all seven players pooled (English, five letters). The median falls as skill rises, since the cost factor does; the lower decile is zero because probes have no daring.

(any word that still fits); *wild* (a common word, ignoring the clues); and *wild-bold* (a rare word, ignoring the clues). Means are over the guesses made with more than one word left. Table 5 gives English at five letters (200 games each); the output of this and three more settings is in `results/daring_calibration.txt`.

Three things stand out. First, the best player has daring exactly 0, as Proposition 8 says. Second, daring separates the players by what they try to do and not by how good they are: the players who go for the word whatever it costs (frequent, random-fit) score about 20; the players who work the clues (near-best, wild) score under 7; best-fit, which goes for the word only when the word is also a good split, scores 4. Third, daring is small for everyone, because the tries-to-spare and chance factors shrink it, so a good player seldom exceeds 10 (Table 6).

The levels depend on the setting. Table 7 gives the means in four settings. The picture is the same in all four: best-fit is at 3.5 to 4.8 (except at eight letters, where it plays almost as well as best, 0.5), and frequent and random-fit are at 14 to 24. Skill depends on length: the player who ignores the clues scores 48 at five letters and 69 at eight, since a longer word has more feedback patterns and a random guess splits better. A skill threshold set at five letters misplaces eight-letter play. The app's thresholds are a first calibration, set at five letters, and not measured on people.

The app turns the two means into a playstyle: skill below 58, from 58 to 85, and 85 or more gives three tiers, and each tier splits by daring at 12, 12 and 2.5 respectively (loose, solid, sharp), so that *bold* means going for the word more than a player of that skill who works the

| | en, 5 | fr, 5 | de, 4 | en, 8 |
|---|---|---|---|---|
| near-best | 99.0 / 0.3 | 98.9 / 0.4 | 97.8 / 0.8 | 99.4 / 0.3 |
| best-fit | 91.4 / 4.0 | 87.1 / 4.8 | 88.7 / 3.5 | 99.0 / 0.5 |
| frequent | 66.3 / 19.7 | 63.7 / 20.7 | 62.6 / 20.2 | 77.1 / 16.7 |
| random-fit | 70.7 / 21.4 | 69.0 / 22.1 | 64.6 / 24.0 | 81.8 / 14.3 |
| wild | 48.2 / 6.0 | 45.3 / 5.5 | 41.5 / 7.3 | 68.9 / 7.0 |

Table 7: Mean skill / mean daring in four settings (200 games each, 100 in English at eight letters). The best player is 100/0 everywhere.

| regressors | en, 5 | fr, 5 | de, 4 |
|---|---|---|---|
| skill | 0.156 | 0.050 | 0.062 |
| luck | 0.286 | 0.425 | 0.490 |
| daring | 0.000 | 0.026 | 0.001 |
| skill + luck | 0.458 | 0.538 | 0.604 |
| skill + daring | 0.215 | 0.153 | 0.112 |
| skill + luck + daring | 0.483 | 0.578 | 0.619 |

Table 8: $R^2$ of the tries a game takes on its mean skill, luck and daring (400 simulated games each; `results/predict_tries.txt`).

clues. This gives six styles. The five-letter bots land as follows: best and near-best are in the sharp, careful style (analyst); best-fit in the sharp, bold one (maverick); frequent and random-fit in the solid, bold one (gambler); wild and wild-bold in the loose, careful one (wanderer). A margin of 3 points of skill and 1.5 of daring around the player's current style keeps one game either side of a line from changing it.

### 9.6 Do skill, luck and daring predict the tries?

If skill, luck and daring describe a game, they should predict how many tries it takes. `app/tools/predict_tries.dart` plays 400 games per language by simulated players whose habits are drawn afresh for each game (the chance, per guess, of each of best, best-fit, frequent, random-fit and wild play), and fits the number of tries (a failed game counts as 7) by least squares on the game's mean skill, mean luck and mean daring over the guesses with a choice. Table 8 gives $R^2$.

Skill and luck together explain about half of the variance, and daring adds 1.5 to 4 points on top ($0.458 \to 0.483$, $0.538 \to 0.578$, $0.604 \to 0.619$), with a negative coefficient in every fit: given skill and luck, a player who went for the word more took fewer tries. Daring alone explains almost nothing, since by itself it does not say whether the attempt paid off. The addition is real but modest; the rest of the variance is the discreteness of the tries and what the game means lose (the number of guesses with a choice, and when the luck came). Luck explains more than skill alone because a game that ends early did so by a lucky split.

The history page uses the fits to draw the expected guesses of a period. It takes the mean of the three fits, $8.05 - 0.0413\,\text{skill} - 0.0272\,\text{luck} - 0.0316\,\text{daring}$, and shows each term as what it does to a base. Skill, which is never negative, can only take guesses off; luck can do either;

daring is measured from a typical level of 8 so that it too can add or take off. The weights come from simulated players and are a first reading, not measured on people.

### 9.7 A worked game

`worked_game.py` follows one game: CARET, then LEAST, against AGENT. After CARET 18 words remain. LEAST splits them into classes of $5, 3, 2, 2, 1, 1, 1, 1, 1, 1$. Its expected pool is $E = 2.67$. The best real guess is SEPAL, with $E^* = 1.56$ and classes $3, 2, 2$ and eleven singletons. LEAST scores 78; SEPAL scores 100.

The answer falls in the class of 5 under LEAST: reduction 1.85 bits = 2.75 (skill) $+0.28$ (risk) $-1.19$ (luck), percentile luck $-72\%$. The best guess SEPAL has skill 3.53 and risk 0.15 bits. LEAST is a possible answer, so it finishes the game at once with chance $1/18 = 5.6\%$; SEPAL is not a possible answer and cannot. This is a gamble taken to finish in fewer guesses. It is priced in the risk term (0.28 against 0.15 bits), but only as a singleton class. The accounting gives a win no more credit than any other singleton, although a win saves a whole guess (Section 10).

### 9.8 The game's answer draw

Luck has mean zero when each possible word is equally likely. The game draws answers by word frequency. `prior_bias.py` gives the mean first-guess luck under that draw: $+10.2\%$ for TARES, $+10.6\%$ for CRANE, $+7.6\%$ for SLATE, $-6.4\%$ for ABBEY, $-8.2\%$ for MAMMA. Restricting the draw to the 500 most common words changes these by about one point. Common words tend to fall in small classes of strong guesses and large classes of weak ones. Under (U) the means are zero to machine precision (Table 3). This is the first open problem.

### 9.9 A bound on the best conceivable split

Three bounds on the number of groups were tested: the naive $\min(N, 3^n)$; a bound $B$ counting 2 outcomes at positions fixed by the clues and 3 elsewhere; and a bound $C$ taking, over all real guesses, the product over letters of how many of green/yellow/grey that letter could show against the words still possible. On 14 simulated games (40 positions after 0, 1 or 2 guesses), $B$ and $C$ equal the naive bound in every position. Where $N > 243$ the best scanned real guess made about 63% of the allowed groups. Where $N \leq 243$ a real guess separated every word in 3 of 20 positions. $C$ is not a valid bound for repeated letters (7 of 109,782 guesses exceeded it, all with a repeated letter). The bounds do not give a reachable best, so skill is normalised by the best real guess.

### 9.10 Numerical check of the theorems

`verify_theorems.py` checks Theorems 3 and 4 and Proposition 5 independently of the app. On 20,000 random partitions, luck lies strictly in $(-1, 1)$, is strictly monotone in the pool, is unchanged by scaling the classes, has $|\text{mean}| < 2\times10^{-16}$, and has variance equal to $(1-\sum q^3)/3$ to the same precision. On 1,500 simulated games of four guesses, with uniform answers and guesses depending only on the past, the mean luck of each guess is within sampling error of zero (for the first guess $+0.008$, standard deviation 0.57). All correlations of a guess's luck

with the skill of any guess so far, and with earlier luck, are within four standard errors of zero (largest in magnitude 0.084, tolerance 0.103).

# 10 Open problems

1. **Non-uniform priors.** For frequency-weighted answers, replace sizes by prior masses: $E = \sum_i W_i^2/W$ with $W_i$ the mass of class $i$. Theorems 3 and 4 hold with $p_i = W_i/W$, which would remove the bias of about $\pm 10$ points measured above. Open: whether the ranked pool is the word count or the mass, and whether a player can be assumed to know the frequencies.

2. **Lookahead.** Skill scores the split made now. A guess can score lower and play better over the rest of the game. Is there a skill that credits lookahead and depends only on the player's knowledge, for example through the expected number of further guesses of an optimal continuation [4]?

3. **Scale of skill.** The logarithm of the expected pool is used because the linear scale compresses first guesses near 100. Other Rényi orders, or a scale calibrated to the guesses a strong player needs, are alternatives. Test against how strong players rank openings.

4. **Risk attitude and tries left.** The best entropy order should depend on the guesses remaining: an even split with many left, a gamble with few (Section 9.4). Is there a skill that depends on the tries left, for example the chance of finishing within them under an optimal continuation?

5. **Credit for finishing now.** A guess that is a possible answer wins at once with chance $1/N$. In the accounting the winning class is a singleton like any other, with reduction $\ln N$. But landing in another singleton still costs one more guess; landing in the winning class does not. Skill and risk therefore undercount guesses that can win. A measure based on the expected number of remaining guesses would count it. How should this be combined with the entropy terms?

6. **Total luck of a game.** Is there a normalised total luck comparable across games of different length, for example $\sum_t \Lambda_t / \sqrt{\sum_t \operatorname{Var} \Lambda_t}$?

7. **Relay-race credit.** How to credit the head start of a board to the earlier guesses that produced it.

8. **Ties.** Mid-ranks are used for equal pools. Strict ranks give a luck with nonzero mean. Other tie rules exist. Does fairness characterise the mid-rank rule?

9. **Hard modes, other feedback.** Only the following is used about $f$: a map to a finite set with all-green exactly at equality. Do the results and the algorithm extend to games where later guesses must obey earlier clues, and how does $E^*$ change?

10. **Playstyles on real players.** The thresholds of Section 9.5 come from simulated players at five letters. They should be set on real play, and probably per length: skill is not comparable across lengths (the player who ignores the clues scores 48 at five letters and 69 at eight). Is there a normalisation of skill and daring, for example by their distribution over a reference set of players, that makes one threshold fit all?

11. **The form of daring.** $D$ is a product of four factors chosen to be monotone in what a bold choice means. Other forms (a sum, other powers, the chance in bits rather than as a probability) would fit the tries better or worse. Which form predicts the tries best, and does a fitted form beat the product?

12. **Approximating $E^*$.** The exact scan is quadratic in the list size. Are there provable shortcuts, such as a lower bound on $E^*$ from letter statistics of $\Omega$, for very large lists or scripts with more patterns?

## 11 Reproducibility

The app computes these quantities in

- `app/lib/logic.dart`: `scoreGuess`, `chanceLuck`, `skillAgainst`, `analyseGuesses`, `splitRisk` and `daringOf` (risk and daring);
- `app/lib/multi_analysis.dart`: `analyseMulti`;
- `app/lib/playstyle.dart`: the tiers, the splits and the margins of Section 9.5.

Tests are in `logic_test.dart`, `review_plots_test.dart`, `multi_analysis_test.dart` and `playstyle_test.dart` under `app/test`: bounds, zero mean over 500 random partitions, the plot geometry against the table, the properties of risk and daring, and where the simulated players of Table 7 fall. The simulated players of Section 9.5 are run with `dart run tools/calibrate_styles.dart [code] [length] [games]` from `app`, which writes to the console; the saved output is `results/daring_calibration.txt`. The tries regression of Section 9.6 is `dart run tools/predict_tries.dart [code] [length] [games]`, saved as `results/predict_tries.txt`. The other experiments are in `docs/paper/experiments`, share `common.py`, and each states its command and output file:

- `first_guess.py`, `first_guess_summary.py`: first guesses;
- `luck_definitions.py`, `prior_bias.py`: luck definitions, effect of the answer draw;
- `play_skill.py`, `risk_attitude.py`: simulated players, the risk question;
- `reduction_accounting.py`, `worked_game.py`: skill, risk and luck terms, and one worked game;
- `capacity_bound.py`: the negative result;
- `verify_theorems.py`: numerical checks.

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